REPORT

1) Explanation of Execution Order (as required)

Using Legacy (P2PKH)

The connection with the docker daemon is already established via bitcoin-cli and we have preconfigured the bitcoin.conf file.

The Info inside the bitcoin.conf file is shown on the right. All the required information is present. The three addresses of A, B and C respectively are shown below:

```
root@8cf81a0bcc50:/# ls -a ~/.bitcoin/bitcoin.conf
/root/.bitcoin/bitcoin.conf
root@8cf81a0bcc50:/# cat ~/.bitcoin/bitcoin.conf
regtest=1
paytxfee=0.0001
fallbackfee=0.0002
mintxfee=0.00001
txconfirmtarget=6
root@8cf81a0bcc50:/#
```

Now we create a transaction between A and B . The part_1_raw file can do this when executed. The transaction ID (A -> B) returned by above program became the input for the 2nd program, whose code is in the part_1_chain file. (this is shown in the second image).

The TXID (A->B) is 588d2955c333402943c26f816f07bf22d24bd3d968d29c9cd20e3971ea1e7498 is the input in our example.

The file part1_chain.cpp executed "listunspent" to find the specific index and "scriptPubKey" associated with txid for (B->C)

```
root@8cf81a0bcc50:/# g++ part1_raw.cpp -o part1_raw
root@8cf81a0bcc50:/# ./part1_raw
--- Part 1: The Setup ---
Address A: mzdNjBBscxsbqopyJzXhzYEyLkuBncMQCB
Address B: mtfLJxg7UPHGxcNg1LsZPaCmGXC7Nrvjiu
Address C: mmYJ5gcRiGdUdSCueg1RuuvYk6iRovCHy8

--- Decoded ScriptPubKey (Challenge) ---
OP_DUP OP_HASH160 902f5732181887494a84a64a4a5c68f72a8219ac OP_EQUALVERIFY OP_CHECKSIG

Final TXID (A -> B): 588d2955c333402943c26f816f07bf22d24bd3d968d29c9cd20e3971ea1e7498
```

Next, the part_1_chain program explicitly referenced the previous txid as its input. This shows how the previous output was used as input for subsequent transaction from B to C, thereby forming a continuous chain of ownership on the regtest blockchain. (Execution output shown in third image).

The decoded script for A->B in our example is:
OP_DUP OP_HASH160 902f5732181887494a84a64a4a5c68f72a8219ac OP_EQUALVERIFY OP_CHECKSIG

In terms of hex, it is **76a914902f5732181887494a84a64a4a5c68f72a8219ac88ac**

```
root@8cf81a0bcc50:/# g++ part1_chain.cpp -o part1_chain
root@8cf81a0bcc50:/# ./part1_chain
--- Program 2 (B -> C Chain) ---
Target Address B: mtfLJxg7UPHGxcNg1LsZPaCmGXC7Nrvjiu

--- Listing UTXO for Address B ---
[
  {
    "txid": "588d2955c333402943c26f816f07bf22d24bd3d968d29c9cd20e3971ea1e7498",
    "vout": 0,
    "address": "mtfLJxg7UPHGxcNg1LsZPaCmGXC7Nrvjiu",
    "label": "B",
    "scriptPubKey": "76a914902f5732181887494a84a64a4a5c68f72a8219ac88ac",
    "amount": 10.00000000,
    "confirmations": 1,
    "spendable": true,
    "solvable": true,
    "desc": "pkh([b85370b2/44h/1h/0h/0/1]039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a)#fda23g3w",
    "parent_descs": [
      "pkh(tpubD6NzVbkrYhZ4XpuEq5UDn9TRh6u7xbABqLrhd4oLdigC4KPdUZBPcDbzkaz7CWXqkywGKLpsaEBv4Ein9Lxqg75M9AADMeKyZehEbCSbxRV/44h/1h/0h/0/*)#qaz92y0s"
    ],
    "safe": true
  }
]

--- Decoded ScriptSig (Response/Unlocking Script) ---
3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c281d4eb64ef237b193899284794dd984983d3cde29b2c585a832d53[ALL] 039c942ea3da439816a4666b174209dbeb7dbd
d677705b9129aa51d1d8ebd0d50a

Final TXID (B -> C): f23af8610461200085f9f7fefe3c9a680f1e392db1949b37e1c26363a619eca2
```

Decoded script for B->C in our example is as follows:

3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c2 8 1d4eb64ef237b193899284794dd984983d3cde29b2c585a832d53[ALL] 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a

in terms of hex, it is 473044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c 281d4eb64ef237b193899284794dd984983d3cde29b2c585a832d530121039c942ea3da439816a4666b 1 74209dbeb7dbdd677705b9129aa51d1d8ebd0d50a

Repeat the process between B and C using getrawtransaction

```
root@8cf81a0bcc50:/# bitcoin-cli getrawtransaction f23af8610461200085f9f7fefe3c9a680f1e392
db1949b37e1c26363a619eca2
020000000198741eea71390ed29c9cd268d9d34bd222bf076f816fc243294033c355298d58000000006a473044
022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c281d4eb64
ef237b193899284794dd984983d3cde29b2c585a832d530121039c942ea3da439816a4666b174209dbeb7dbdd6
77705b9129aa51d1d8ebd0d50afdffffff020065cd1d000000001976a9144211b012c22c798776dbe75da58535
c91af5cf2a88ac60decb1d000000001976a914902f5732181887494a84a64a4a5c68f72a8219ac88ac00000000
root@8cf81a0bcc50:/# bitcoin-cli sendrawtransaction "020000000198741eea71390ed29c9cd268d9d
34bd222bf076f816fc243294033c355298d58000000006a473044022028e82ff4424515e741f608b0a3c5bae40
69621a64ff546f17caa18ff8c82d861022005dd4471c281d4eb64ef237b193899284794dd984983d3cde29b2c5
85a832d530121039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50afdffffff020
065cd1d000000001976a9144211b012c22c798776dbe75da58535c91af5cf2a88ac60decb1d000000001976a91
4902f5732181887494a84a64a4a5c68f72a8219ac88ac00000000"
f23af8610461200085f9f7fefe3c9a680f1e392db1949b37e1c26363a619eca2
root@8cf81a0bcc50:/# bitcoin-cli generatetoaddress 1 $(bitcoin-cli getnewaddress)
[
  "40e15759549331750be0cbad7012dcfe673ae49628ab78d4696de45165e7a064"
]
```

Following is the list of UTXOs as a result of these transactions (using "listunspent"):

```
root@8cf81a0bcc50:/# bitcoin-cli listunspent
[
  {
    "txid": "f23af8610461200085f9f7fefe3c9a680f1e392db1949b37e1c26363a619eca2",
    "vout": 0,
    "address": "mmYJ5gcRiGdUdSCueg1RuuvYk6iRovCHy8",
    "label": "C",
    "scriptPubKey": "76a9144211b012c22c798776dbe75da58535c91af5cf2a88ac",
    "amount": 5.00000000,
    "confirmations": 1,
    "spendable": true,
    "solvable": true,
    "desc": "pkh([b85370b2/44h/1h/0h/0/2]02fd2b6671c5fa09a70b3bb23e4c6a0cab0ca9f306c38f70450d5346171761a03d)#w3kxsh8z",
    "parent_descs": [
      "pkh(tpubD6NzVbkrYhZ4XpuEq5UDn9TRh6u7xbABqLrhd4oLdigC4KPdUZBPcDbzkaz7CWXqkywGKLpsaEBv4Ein9Lxqg75M9AADMeKyZehEbCSbxRV/44h/1h/0h/0/*)#qaz92y0s"
    ],
    "safe": true
  },
  {
    "txid": "f23af8610461200085f9f7fefe3c9a680f1e392db1949b37e1c26363a619eca2",
    "vout": 1,
    "address": "mtfLJxg7UPHGxcNg1LsZPaCmGXC7Nrvjiu",
    "label": "B",
    "scriptPubKey": "76a914902f5732181887494a84a64a4a5c68f72a8219ac88ac",
    "amount": 4.99900000,
    "confirmations": 1,
    "spendable": true,
    "solvable": true,
    "desc": "pkh([b85370b2/44h/1h/0h/0/1]039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a)#fda23g3w",
    "parent_descs": [
      "pkh(tpubD6NzVbkrYhZ4XpuEq5UDn9TRh6u7xbABqLrhd4oLdigC4KPdUZBPcDbzkaz7CWXqkywGKLpsaEBv4Ein9Lxqg75M9AADMeKyZehEbCSbxRV/44h/1h/0h/0/*)#qaz92y0s"
    ],
    "safe": true
  },
```

Here In Our Example :

5 BTC goes to C and as a result 4.999 BTC goes back to B(which was initially 10 BTC). This was because a transaction fee of 0.001 BTC was allocated in the regtest environment.

To check if the if the response matches with the challenge script provided by the transaction, we took the response script which was 47 units long (reason why we added the number 47 as prefix before the response script as other methods didn't work). And we replace the [ALL] part with 01 and then the part starting from 0368d4… is prefixed with it's length (21) and then the scriptPubKey (which is the challenge script) is appended at the end.

These prefixes are data push opcodes. 0x47 means to push the next 71 bytes (the signature). 0x21 means to push the next 33 bytes(the public keyStructure of challenge, response and how validation occurs:

```
root@8cf81a0bcc50:/# btcdeb 473044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c281d4eb64ef237b193899284794dd984983d3cde29b2c585a832d53012 1039c942ea3
da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a76a914902f5732181887494a84a64a4a5c68f72a8219ac88ac
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
7 op script loaded. type `help` for usage information
script                                                                  |  stack
------------------------------------------------------------------------+--------
3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...      |
039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a      |
OP_DUP                                                                  |
OP_HASH160                                                              |
902f5732181887494a84a64a4a5c68f72a8219ac                                |
OP_EQUALVERIFY                                                          |
OP_CHECKSIG                                                             |
#0000 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c281d4eb64ef237b193899284794dd984983d3cde29b2c585a832d5301
btcdeb> _
```

The transaction uses P2PKH. The response (ScriptSig) contains the digital signature and sender's public key. The challenge (scriptPubKey) contains the cryptographic instructions and the target hash.
We type step and this pushes the signature to the stack

```
btcdeb> step
                <> PUSH stack 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18ff8c82d861022005dd4471c281d4eb64ef237b193899284794dd984983d3cde29b2c585a832d5301
script                                                                  |                                                                  stack
------------------------------------------------------------------------+-------------------------------------------------------------------
039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a      | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
OP_DUP                                                                  |
OP_HASH160                                                              |
902f5732181887494a84a64a4a5c68f72a8219ac                                |
OP_EQUALVERIFY                                                          |
OP_CHECKSIG                                                             |
#0001 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
```

We again type step and this pushes the public key to the stack

```
btcdeb> step
                <> PUSH stack 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
script                                                                  |                                                                  stack
------------------------------------------------------------------------+-------------------------------------------------------------------
OP_DUP                                                                  | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
OP_HASH160                                                              | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
902f5732181887494a84a64a4a5c68f72a8219ac                                |
OP_EQUALVERIFY                                                          |
OP_CHECKSIG                                                             |
#0002 OP_DUP
```

executing step the third time executes the OP_DUP.

```
btcdeb> step
                <> PUSH stack 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
OP_HASH160                                                      | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
902f5732181887494a84a64a4a5c68f72a8219ac                        | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
OP_EQUALVERIFY                                                  | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
OP_CHECKSIG                                                     |
#0003 OP_HASH160
```

The next step executes the OP_HASH160. It takes the public key provided in response and hashes it. OP_EQUALVERIFY then checks if this result matches the challenge. This hash matches the Address B from 1st Program. And thus proves that the public key belongs to address B.

```
btcdeb> step
                <> POP  stack
                <> PUSH stack 902f5732181887494a84a64a4a5c68f72a8219ac
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
902f5732181887494a84a64a4a5c68f72a8219ac                        |                                 902f5732181887494a84a64a4a5c68f72a8219ac
OP_EQUALVERIFY                                                  | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
OP_CHECKSIG                                                     | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
#0004 902f5732181887494a84a64a4a5c68f72a8219ac
```

The fifth step pushes the target hash from the script onto the stack

```
btcdeb> step
                <> PUSH stack 902f5732181887494a84a64a4a5c68f72a8219ac
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
OP_EQUALVERIFY                                                  |                                 902f5732181887494a84a64a4a5c68f72a8219ac
OP_CHECKSIG                                                     |                                 902f5732181887494a84a64a4a5c68f72a8219ac
                                                                | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
                                                                | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
#0005 OP_EQUALVERIFY
```

The sixth step executes OP_EQUALVERIFY

```
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
                <> POP  stack
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
OP_CHECKSIG                                                     | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
                                                                | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
#0006 OP_CHECKSIG
```

OP_CHECKSIG verifies that the signature was created by the private key corresponding to the public key. This proves that the owner of the address actually authorized the move to address C.

The 4th step shows the correctness since the hash matches with hash of address B returned by 1st program.

```
btcdeb> step
                <> POP  stack
                <> PUSH stack 902f5732181887494a84a64a4a5c68f72a8219ac
script                                                          |                                                                  stack
----------------------------------------------------------------+------------------------------------------------------------------
902f5732181887494a84a64a4a5c68f72a8219ac                        |                                 902f5732181887494a84a64a4a5c68f72a8219ac
OP_EQUALVERIFY                                                  | 039c942ea3da439816a4666b174209dbeb7dbdd677705b9129aa51d1d8ebd0d50a
OP_CHECKSIG                                                     | 3044022028e82ff4424515e741f608b0a3c5bae4069621a64ff546f17caa18f...
#0004 902f5732181887494a84a64a4a5c68f72a8219ac
```